Factors (Refer to MAS pages15-27 to make notes on this section)

TYPE 1 ; Highest Common Factor (HCF)

Factorise: $12a^2b + 16ab$

Solution: $4ab(3a + 4)$

Taking out a Common Bracket

Factorise: $2a(x + y) - 3b(x + y)$

Solution: $(x+y)(2a - 3b)$

Refer to the examples at the bottom of page 16 and make notes on **changing the sign** before a bracket before taking out the common bracket:

Exercise 1

Factorise the following expressions, by taking out the common factor

1.

a. $40x+20y$ $20(2x+y)$

b. $9d-3e$ $3(3d-e)$

c. $15x-10y$ $5(3x-2y)$

d. $-4x+16y$ $-4(x-4y)$

e. $24d^2-3de$ $3d(8d-e)$

f. $14x^3-49xy$ $7x(2x^2-7y)$

g. $18x^2+45x$ $9x(2x+5)$

h. $35m-14m^2$ $7m(5-2m)$

i. x^4-x^3 $x^3(x-1)$

j. x^5-x^3 $x^3(x^2-1)$ $x^3(x+1)(x-1)$

k. $-xy+x^2$ $-x(y-x)$

l. $\frac{1}{2}x-1$ $\frac{1}{2}(x-2)$

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$2ap(2x+ap^2y)$ $33d(3d-e)$ $xy(16x-y)$

a. $4apx+2a^2p^3y$ b. $99d^2-33de$ c. $16x^2y-xy^2$

d. $4x+11y$ no factors e. $\pi x^2+\pi x$ $\pi x(x+1)$ f. $54-9$ $9(6-1)=9\times5=45$

g. $\frac{1}{2}m-8m^2$ $\frac{1}{2}m(1-16m)$ h. $3^2x^5-3x^3$ i. 5^2-5 $5(5-1)=5(4)$

j. $\frac{1}{2}x-\frac{1}{4}$ $\frac{1}{4}(2x-1)$ $3x^3(3x^2-1)$

$\frac{1}{2}(x-\frac{1}{2})$ $(x+1)(a+7)$

a. $2(x+1)+7(x+1)$ $9(x+1)$ b. $a(b+3)-2(b+3)$ $(b+3)(a-2)$

c. $3(2x+1)+5(2x+1)$ $(2x+1)(8)$ d. $p(q+9)-2(q+9)$ $(q+9)(p-2)$

e. $2m(n-3)+3n(n-3)$ $(n-3)(2m+3n)$ f. $2a(a-b)-2(a-b)$ $(a-b)(2a-2)$ $2(a-1)(a-b)$

a. $2(x-1)-3(x-1)$ $(x-1)(-1)$ b. $a(b+3)-2(b+3)$ $(b+3)(a-2)$

c. $3(x+y)+2(x+y)$ $(x+y)5$ d. $p(p+2)-2(p+2)$ $(p+2)(p-2)$

e. $2m(n-3)+(n-3)$ $(n-3)(2m+1)$ f. $a(a-b)-(a-b)$ $(a-b)(a-1)$

ercise 2: MAS page 17 Exercise 3 No 2

TYPE 2 ; Factorising a Difference of two squares

$A^2 - B^2 =$ _____

Refer to page 17 and make notes on more examples:

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Exercise 3

1. Factorise the following expressions, by using the difference of two squares.

(5d-6f)(5d+6f)

a. $25d^2 - 36f^2$

b. $w^2y^2 - 1$ (wy+1)(wy-1)

c. $4x^2y^2 - 9$ (2xy+3)(2xy-3)

$(d^2-3f^2)(d^2+3f^2)$

d. $d^4 - 9f^4$

e. $4w^2 - 25y^2$ (2w-5y)(2w+5y)

f. $4x^2 + y^2$ *(ouch!)*

no factors

g. $100r^2 + v^2$

h. $36 - y^2z^2$ (6-yz)(6+yz)

i. $x^2y^2 - b^2c^2$ (ay-bc)(xy+bc)

j. $\frac{1}{4}a^2 - 1$ $(\frac{1}{2}a+1)(\frac{1}{2}a-1)$

k. $x^2 - \frac{1}{9}$ $(x-\frac{1}{3})(x+\frac{1}{3})$

l. $2x^4 - 8$ $2(x^4-4)$ $2(x^2+2)(x^2-2)$

2. Factorise the following:

a. $d^2 - f^2$ (d+f)(d-f)

b. $w^2 - z^2$ (w+z)(w-z)

c. $4x^2 - y^2$ (2x+y)(2x-y)

d. $16d^2 - 9f^2$ (4d-3f)(4d+3f)

e. $4w^2 - 25y^2$ (2w+5y)(2w-5y)

f. $49x^2 - 121y^2$ (7x-11y)(7x+11y)

g. $100r^2 - v^2$ (10r+v)(10r-v)

h. $144w^2 - z^2$ (12w+z)(12w-z)

i. $4x^2 - 9y^2$ (2x-3y)(2x+3y)

j. $a^4 - b^2$ $(a^2+b)(a^2-b)$

k. $w^4 - 4z^2$ $(w^2-2z)(w^2+2z)$

l. $169x^2 - 196y^2$ (13x-14y)(13x+14y)

Exercise 4

1. Factorise the following completely:

3(d-f)(d+f)

a. $3d^2 - 3f^2$

(3w+z)(3w-z)

b. $9w^2 - z^2$

(2xy+3)(2xy-3)

c. $4x^2y^2 - 9$

a(ad-f)(ad+f)

d. $a^3d^2 - af^2$

5(x-5y)(x+5y)

e. $5x^2 - 125y^2$

2p(x-3y)(x+3y)

f. $2px^2 - 18py^2$

a(10r-v)(10r+v)

g. $100ar^2 - av^2$

3q(w²+4z²)

h. $3qw^2 + 12qz^2$

2b(25a²-1)

i. $50a^2b - 2b$

$b^2(a^2+1)(a+1)(a-1)$

j. $a^4b^2 - b^2$

$(58+57)(58-57) = 115$

k. $58^2 - 57^2$

2b(5a+1)(5a-1)

l. $16x^2 - 1$

(4x+1)(4x-1)

2xy(6x+1)(6x-1)

m. $72x^3y - 2xy$

(x-1)(4-3y)(4+3y)

n. $16(x-1) - 9y^2(x-1)$

MORE COMPLEX EXAMPLES:

1) $1-16p^{16}$

$=(1+4p^8)(1-4p^8)$

$=(1+4p^8)(1+2p^4)(1-2p^4)$

2) Using substitution:

$1-(x-2)^2$

let $(x-2)=k$

$1-k^2$

$=(1+k)(1-k)$

$=(1+x-2)(1-(x-2))$

$=(-1+x)(1-x+2)$

$=(x-1)(3-x)$

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c) More substitution:

$4(x-3)^2-49(x+1)^2$

let $(x-3)=k$ and $(x+1)=m$

$4k^2-49m^2$

$=(2k+7m)(2k-7m)$

$=[2(x-3)+7(x+1)][2(x-3)-7(x+1)]$

$=[2x-6+7x+7][2x-6-7x-7]$

$=[9x+1][-5x-13]$

2. Now try these:.

a. *Factorise*: x^2-1

hence factorise: $(a-b)^2-1$

b. *Factorise*: x^2-y^2

hence *factorise*: $(a-2)^2-(a+1)^2$

c. *Factorise*: ax^2-4ay^2

hence *factorise*: $a(b+3)^2-4a(b-3)^2$

2a $(x+1)(x-1)$

b) $(a-b+1)(a-b-1)$

b) $(x+y)(x-y)$

$((a-2)+(a+1))((a-2)-(a+1))$

$(2a-1)(-3)$

c) $a(x^2-4y^2)$

$a(x+2y)(x-2y)$

$a((b+3)+2(b-3))((b+3)-2(b-3))$

$a(b+3+2b-6)(b+3-2b+6)$

$a(3b-3)(-b+9)$

$3a(b-1)(-b+9)$

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